

BIODICAPT: Optimal Experimental Design

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Introduction

OED

Optimal Experimental Design (or OED for short¹) is a method for designing experiments that aims at maximising information gain while minimising a cost function. This cost function can encompass uncertainty, cost, time... it is up to the user to select the parameters to optimise.

The key concept is that **not all observations are equally informative**.

This concept can be particularly interesting when working in participatory science, which is our case for the BIODICAPT project.

BIODICAPT

BIODICAPT is a French initiative that aims at monitoring biodiversity of agricultural lands on a large (*aka national*) scale through the use of various recording devices.

The project consists of two phases:

1. Data collection within a “research” network of agricultural plots.
2. Data collection within the 500 ENI network.

500 ENI is a network of farmers that can be recruited to work on BIODICAPT. However, their participation is *voluntary*, which raises some challenges. Therefore, the transition from the first phase to the second involves a significant shift in scale and philosophy for data collection, which is why OED will be needed.

Strategy

During the first phase of the project, data will be collected on ~125 agricultural plots. With our budget and logistical constraints, we aim at sampling 50 to 150 new plots within 500 ENI.

With OED, the question we want to answer is simple: **which points are most informative in the 500ENI network, given our previous data collection and our constraints?**

For that, we'll use explicative variables collected on site (temperatures, precipitation, type of land, ...), response variables obtained from recorders (presence-absence of birds species mainly), and Species Distribution Models (SDM), in particular HMSC, which is a joint-SDM (JSDM) [1]. These models make use of several species distributions to find *latent* links between them, which supposedly improves the prediction accuracy.

They are opposed to single SDMs (Species Distribution Models applied to one species) and SSDMs (Stacked SDMs, are multiple SMDs glued together, with no link between them).

HMSC

HMSC (Hierarchical Modelling of Species Communities) is a JSDM developed by Ovaskainen et al. (2017) [1], it is distributed as an open-source R-package on [GitHub](#) [2].

It is undoubtedly the most widely used JSDM in the last years. To summarise its principle, take a quick look at Figure 4, from their original publication [1]:

¹Sometimes, this concept is also referred to as “adaptive sampling”.

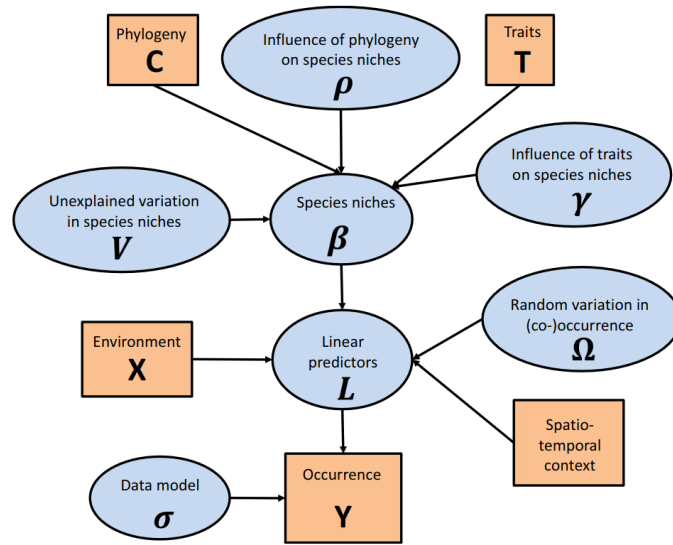


Figure 4 A graphical summary of the HMSC statistical framework. In this Directed Acyclic Graph (DAG), the orange boxes refer to data, the blue ellipses to parameters to be estimated, and the arrows to functional relationships described with the help of statistical distributions.

Figure 1: Copy of Figure 4 from Ovaskainen et al. (2017) [1]

GJAM

🔥 Work in progress...

Only HMSC has been implemented into our code for now. GJAM is under consideration, as well as other SDMs: RF, SSDM, deep-SDMs, ...
This section will be updated when results are available.

Proof-of-concept

Materials & Methods

The results from the first phase of the BIODICAPT data collection will not be available before early 2027. In the meantime, we can use another (but similar) dataset, already curated and known to work with HMSC: STOC [3].

The French Breeding Bird Survey (FBBS, or STOC in french) is a standardised multi-species abundance dataset, based on a citizen science program. It contains bird sightings and explanatory/environmental variables as the BIODICAPT dataset will. However, as it is already pre-processed, we can work directly with it, out of the box.

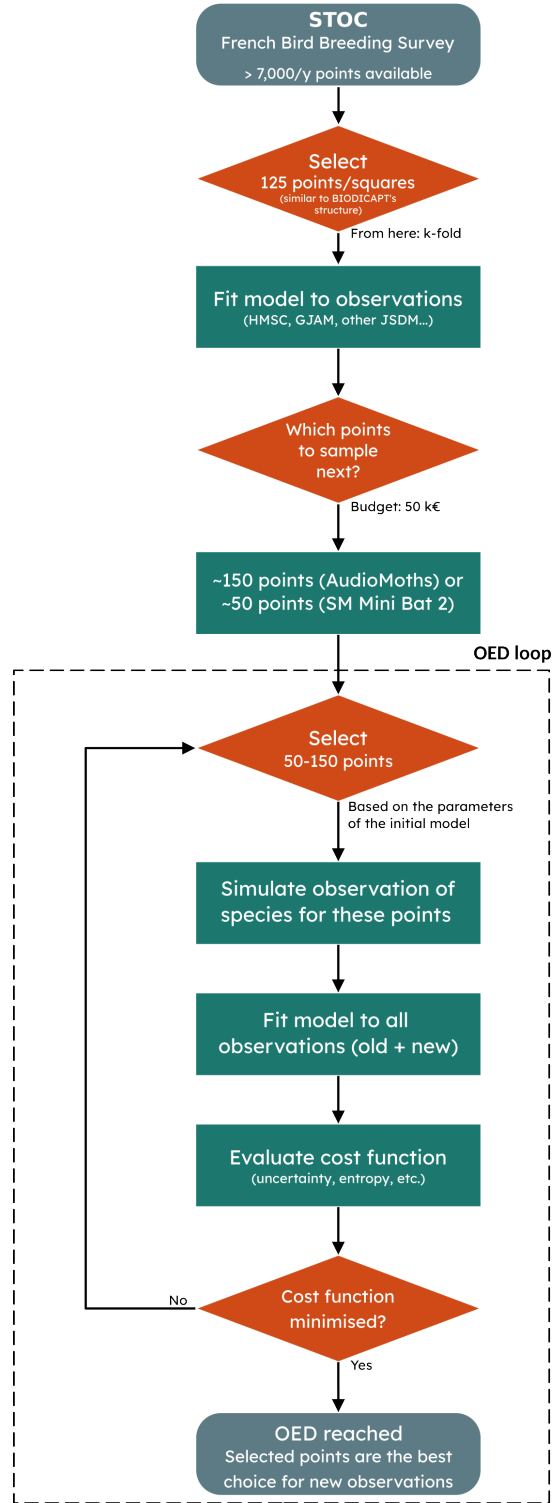


Figure 2: Flowchart of the proof of concept conducted with STOC dataset

Results compared

First, we want to make sure that, following the [flowchart](#) above, we can actually make better models by adding (50-150) new points to our dataset.

Reference results

To test this, we draw on *Mondain-Monval et al.* work [4]. However, it is only a base on which we build, not an exact replica, as our method is quite different from the original *Mondain-Monval et al.* publication:

Parameter	M.M.	Ours
Dataset	Simulated with <code>virtualspecies</code> package	STOC annual sampling
Number of species	50 species	5 species
Number of variables	10 (randomly selected from 33 available)	6 (direct from STOC)
Model type	Ensemble model (GLM + GAM + RF)	HMSC
Re-training strategy	Re-sampling of 1%, 10% or 50% of dataset	Addition of new samples
Training size	2000 cells	125 cells + new data

With this context in mind, here are their results:

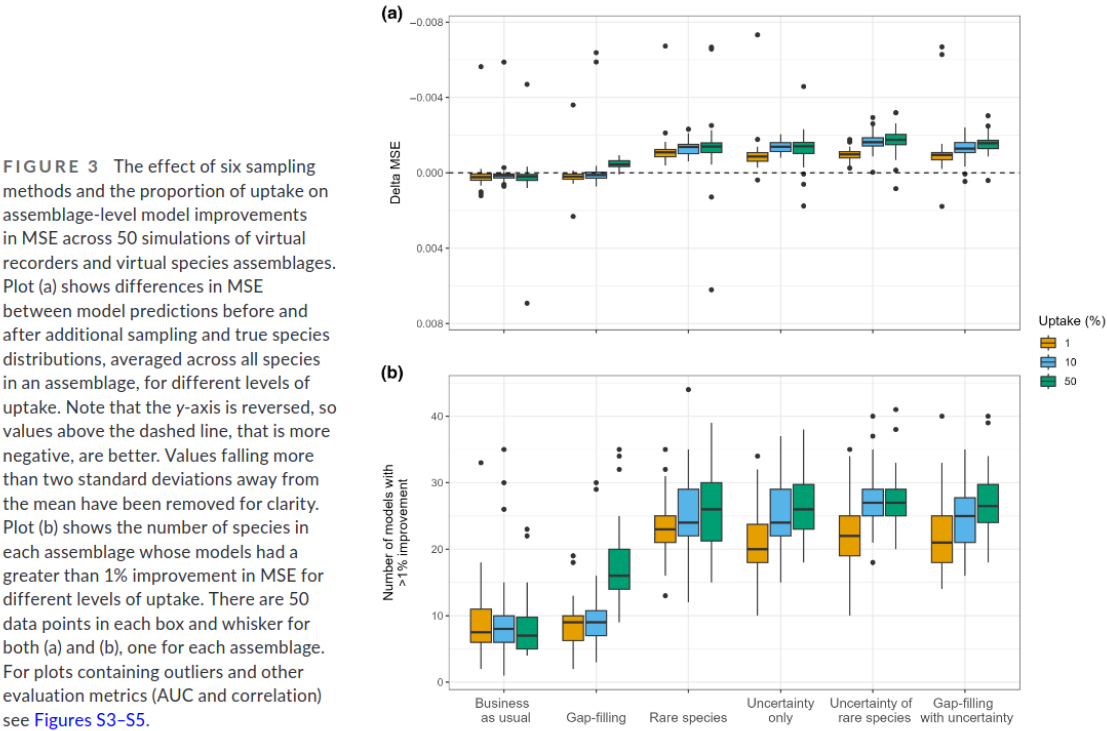
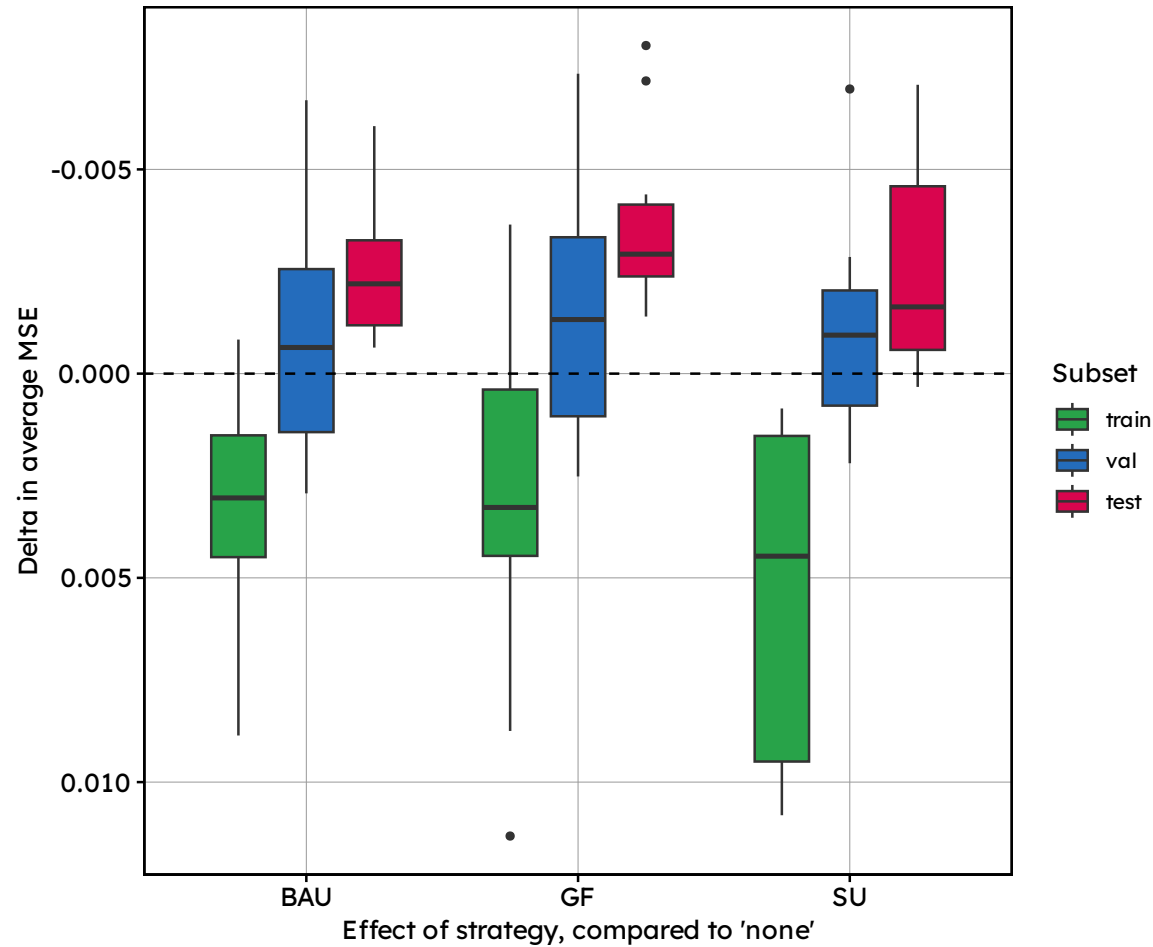


Figure 3: Copy of Figure 3a mondain-monval et al., 2024

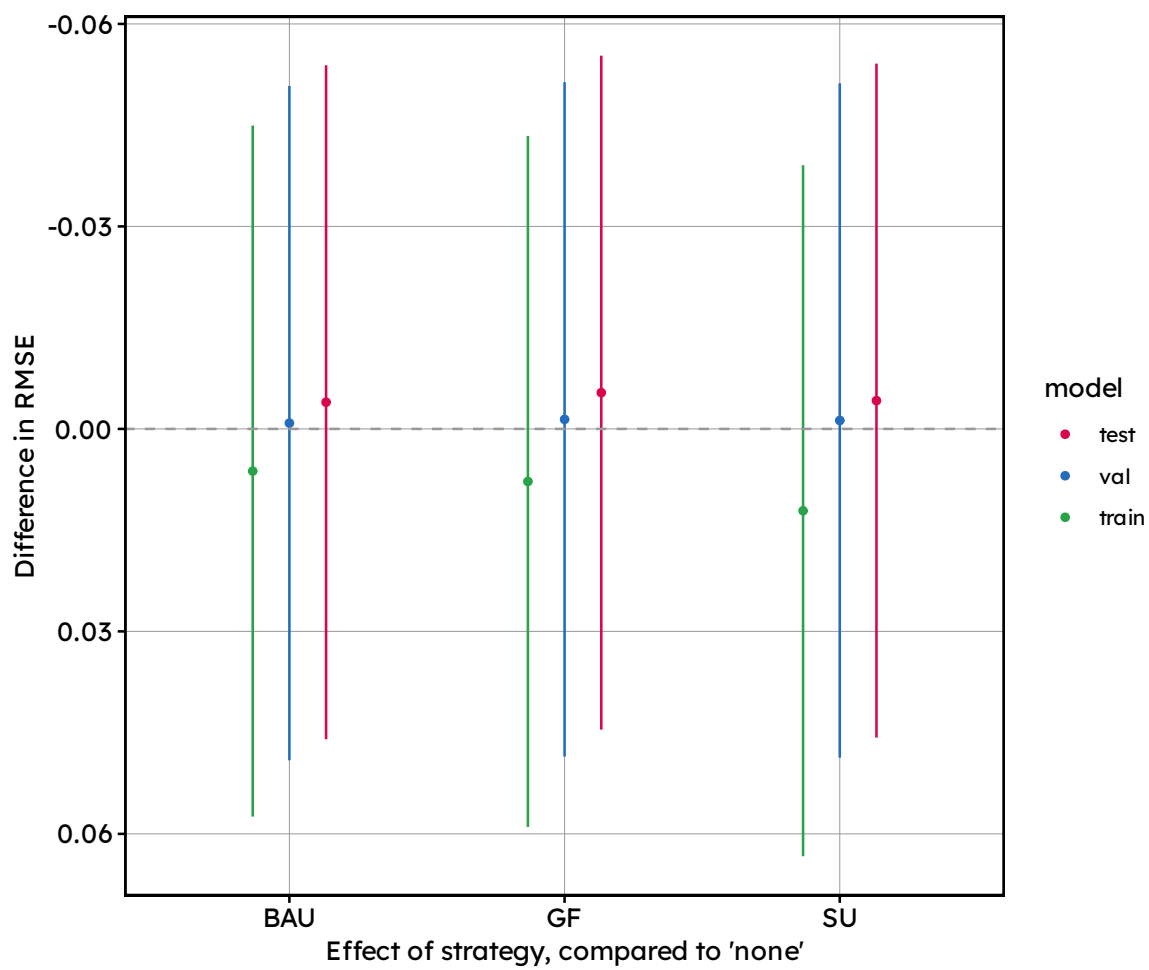
Our results

Boxplot differences



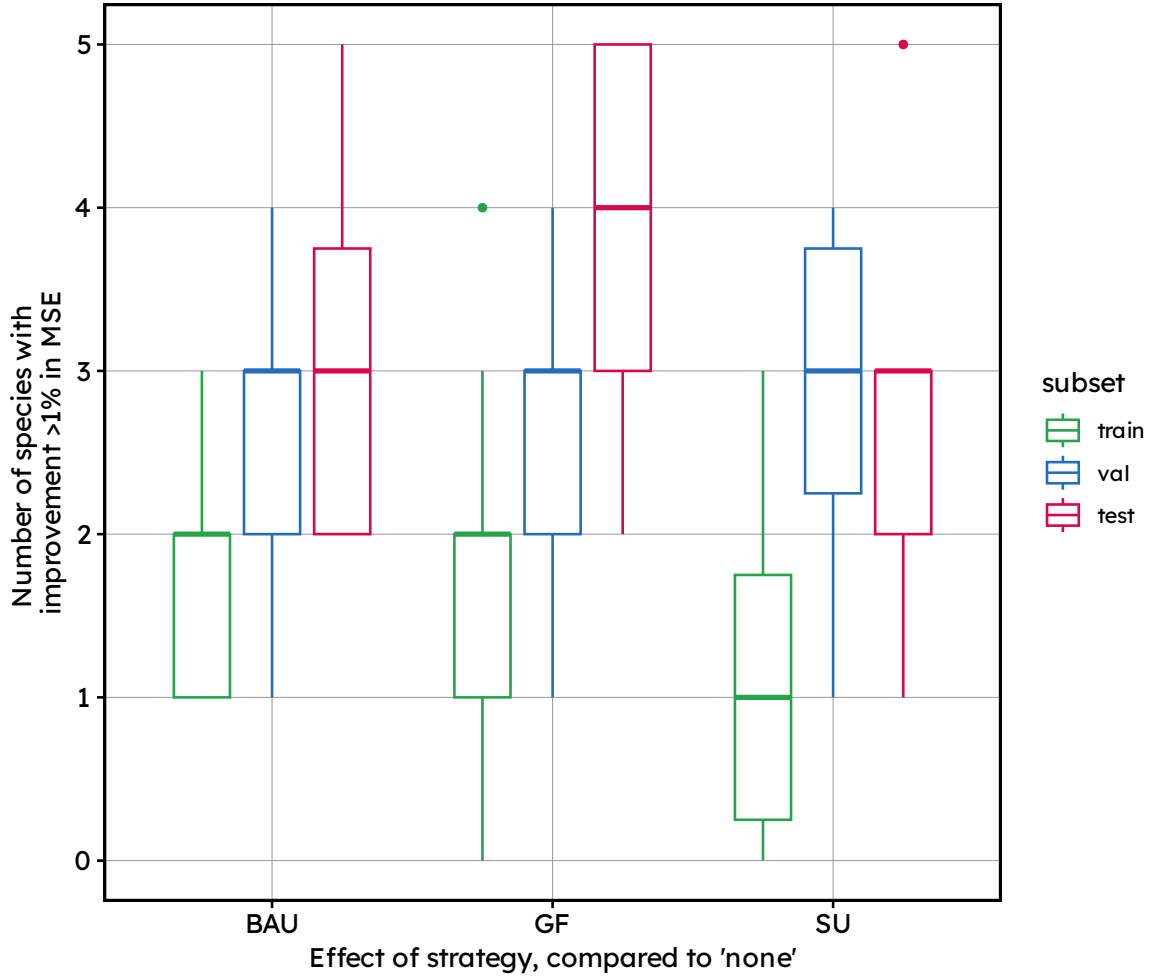
Distribution of per-fold differences (species-averaged within each fold, across k_fold)

Dotwhisker differences



95% CI of scores, average per k_fold.

Boxplot improvement



Number of species per box: 5 .

BAU: *Business-as-usual* ; GF: *Gap-filling* ; SU: *Simplified-uncertainty*

Discussion of comparison

What can we see?

Mondain-Monval et al. [4] show results in boxplots, for several proportion of their modified. We show results for the same addition of training points, but divided into train/val/test subsets.

In our results, differences in MSE between resampling strategies are slight, less marked than those of [4]. However, we note that the “BAU” strategy yields to a more visible decrease in MSE (aka decrease of errors).

It is also a surprise to note that the strategy “SU” does not yield to the best results (which was what we expected from [4]). When looking at the boxplots for the training scores of “SU”, an explanation could be that the model was confronted to much variability (compared other training subsets), which as a slightly negative impact on predictions in test. This is actually quite logical: the “SU” strategy adds training samples for which the predictions of the model are the most uncertain, which is why variability is wider.

What to conclude?

Subset comparisons confirm that training is harder with new unknown samples, but validation

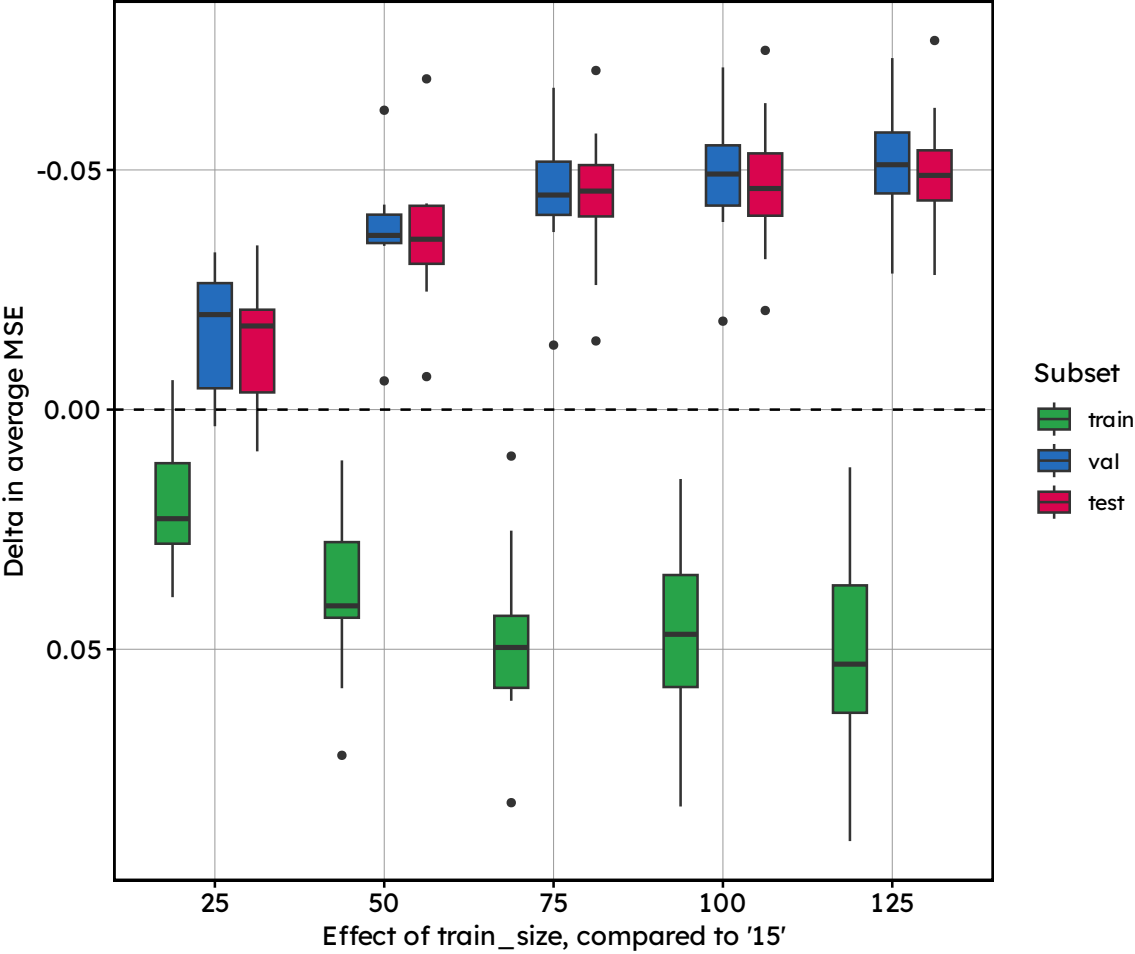
scores are better, as well as test scores. However, the differences with the base strategy is not statistically significant.

Overall, the addition of new points in the sampling strategies has slight a positive effect on the predictions of the models on the test set. The best performing strategy on our practical dataset, using HMSC, seems to be “gap-filling” (GF).

Other results

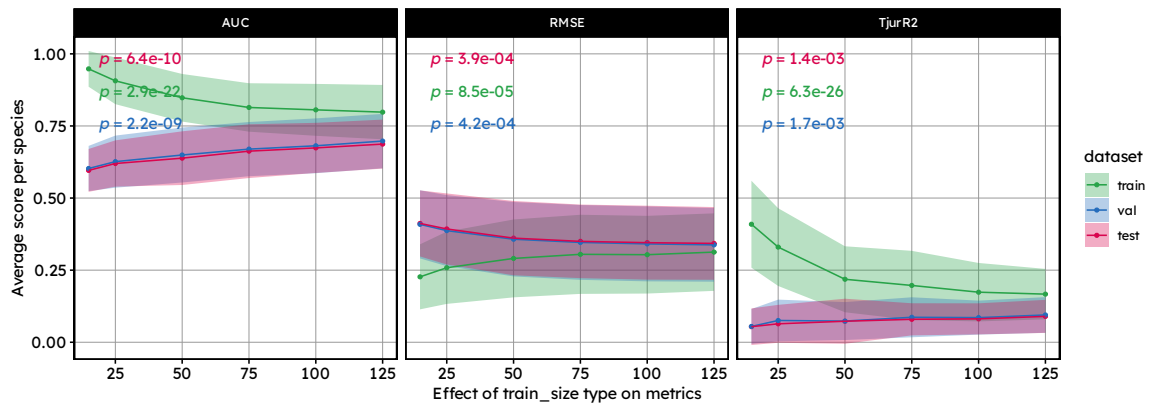
Training size

Boxplot differences



Distribution of per-fold differences (species-averaged within each fold, across k_fold)

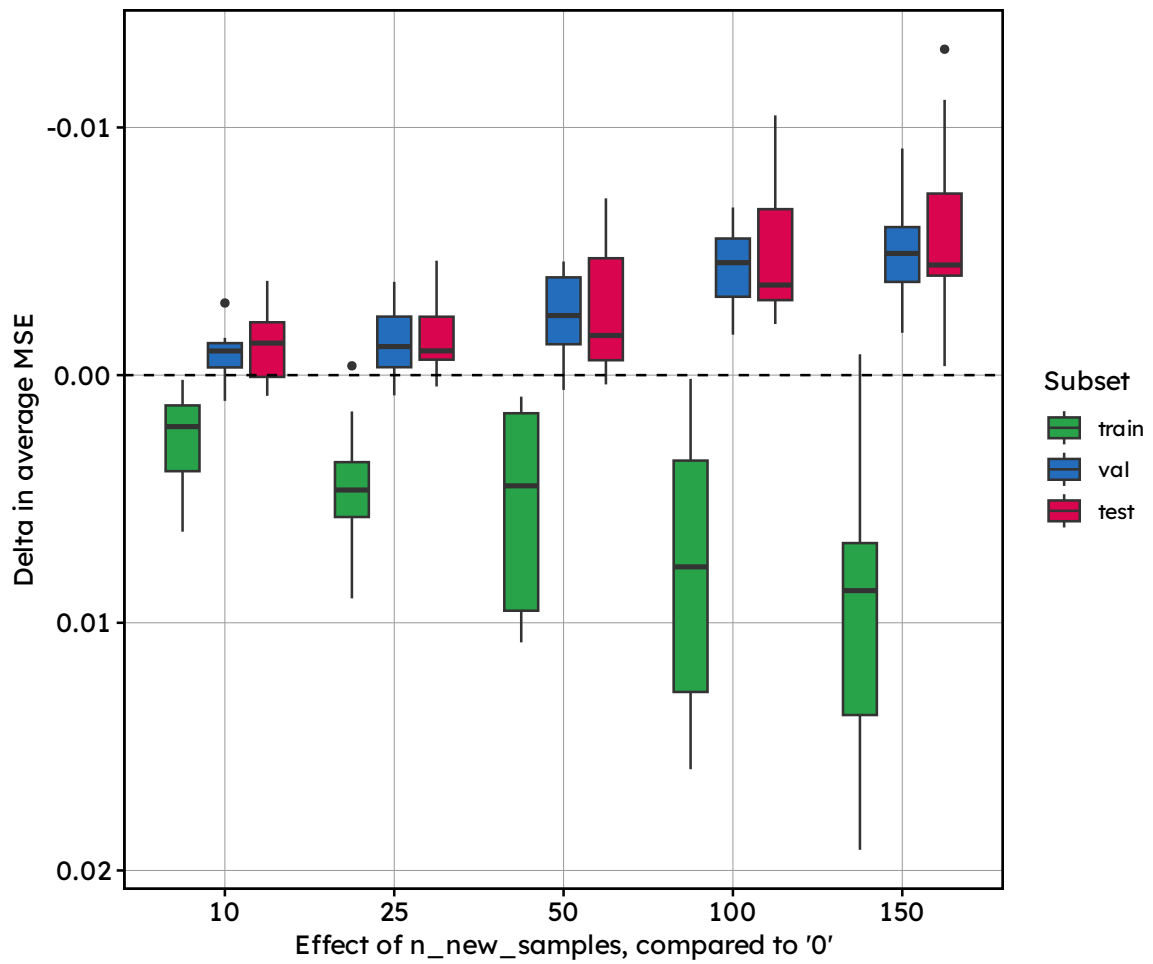
Lineplot



SD and mean computed per k_fold (over all species)

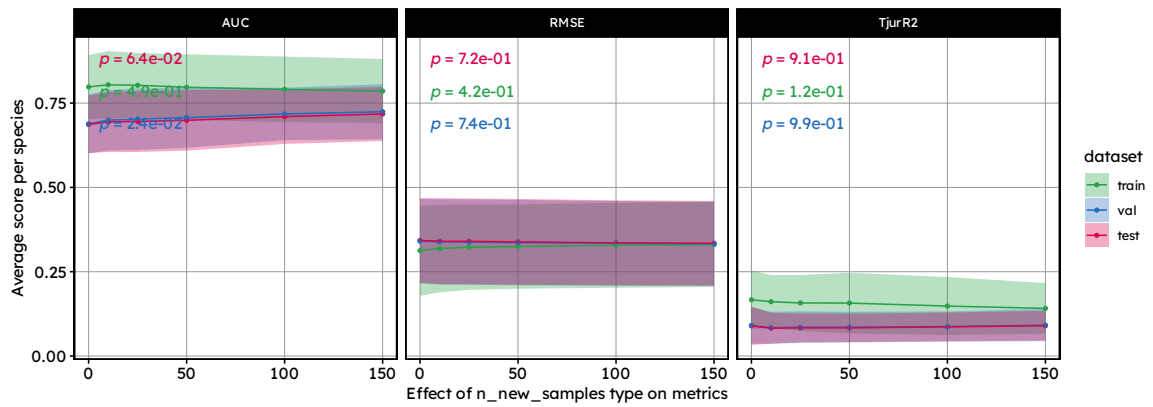
Number of new samples

Boxplot differences



Distribution of per-fold differences (species-averaged within each fold, across k_fold)

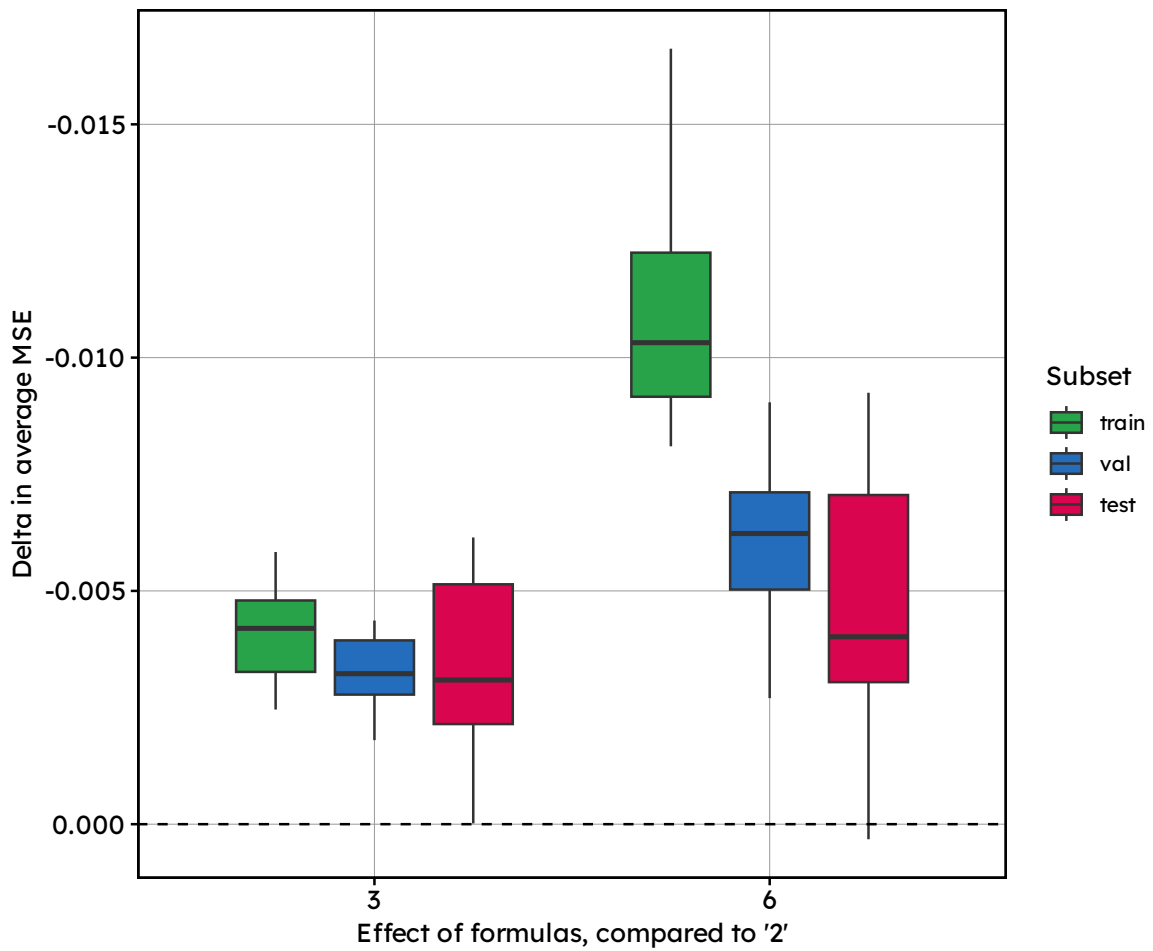
Lineplot



SD and mean computed per k_fold (over all species)

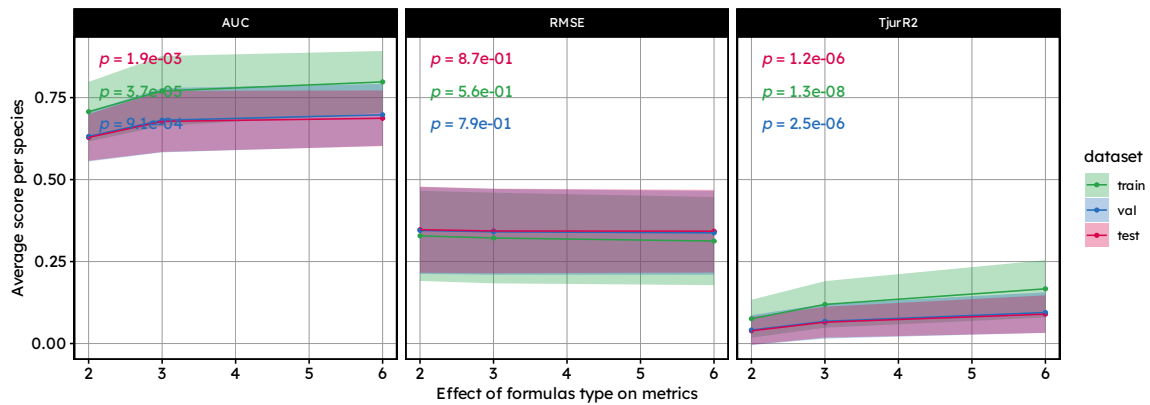
Number of explicative variables

Boxplot differences



Distribution of per-fold differences (species-averaged within each fold, across k_fold)

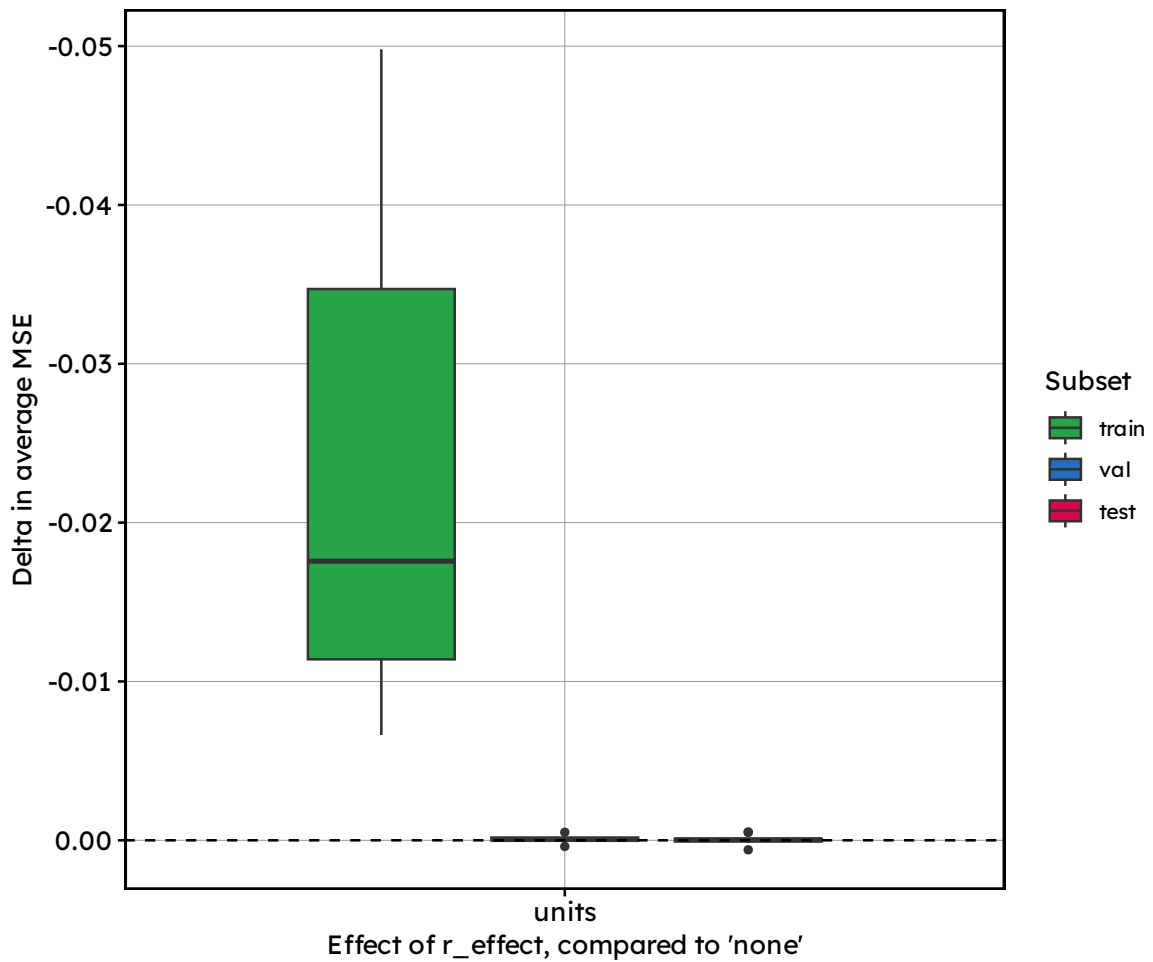
Lineplot



SD and mean computed per k_fold (over all species)

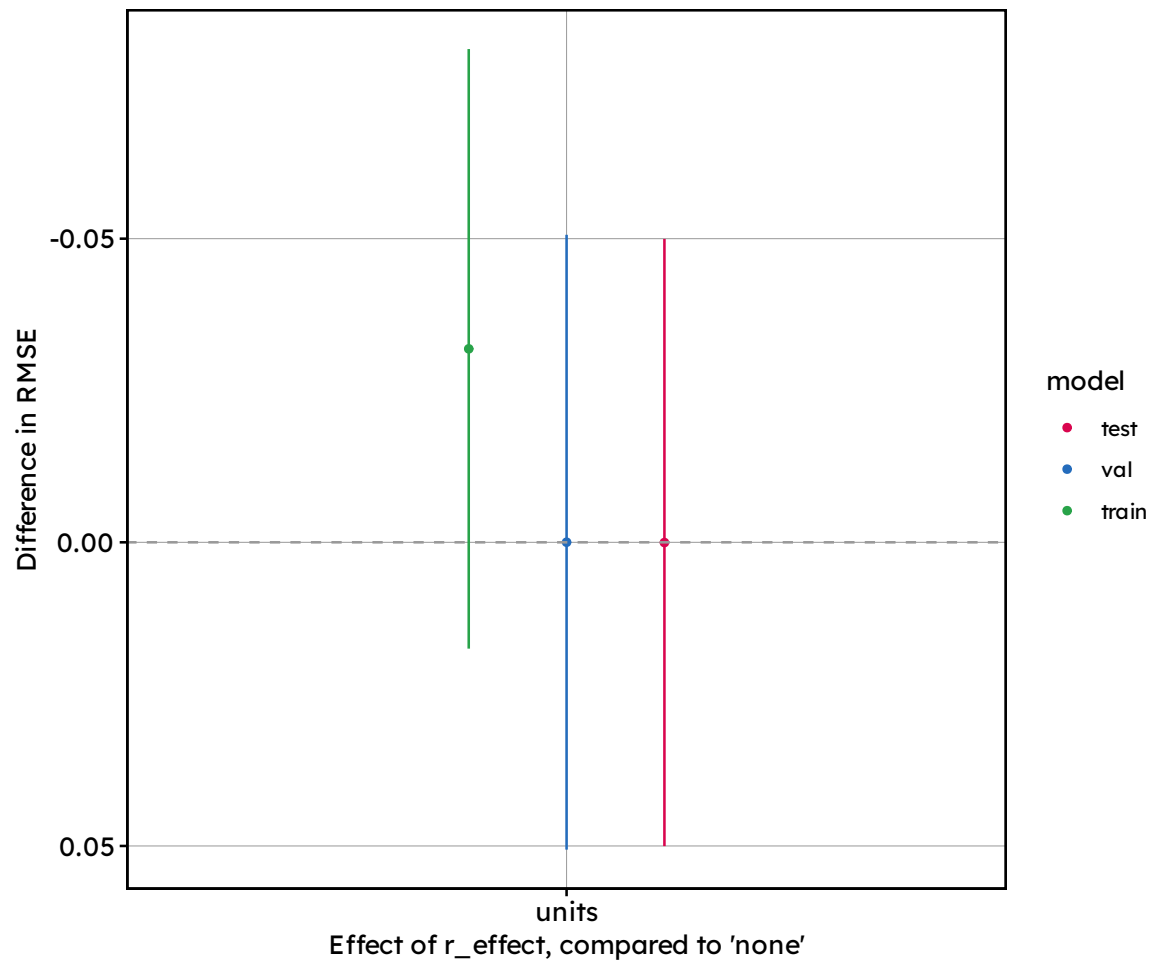
Random Effect

Boxplot differences



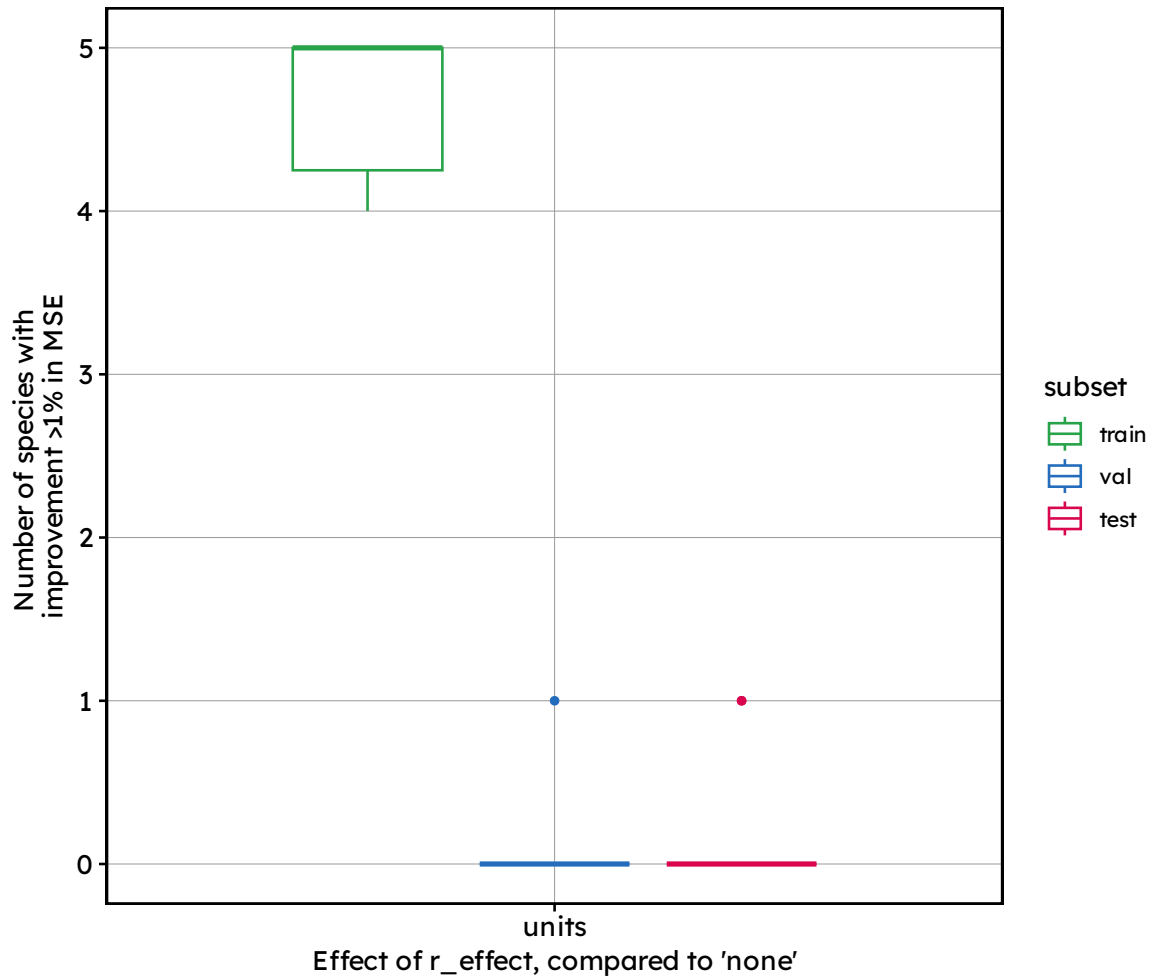
Distribution of per-fold differences (species-averaged within each fold, across k_fold)

Dotwhisker differences



95% CI of scores, average per k_fold.

Boxplot improvement



Number of species per box: 5 .

Discussion of other results

Looking at the other plots : - Training size as a significant (all p-values < 0.001) effect on scores. - Number of new samples does not seem to have any significant effect (which is weird). - Number of explicative variables as an effect (all p-values < 0.001). - Adding a random effect to HMSC yields to better scores in training but no differences in val or test.

Overall, the larger the number of training samples, explicative variables and/or new samples, the better the model performs in val/test, and the worst it performs in training. Although the effects are small, they are comparable to the ones of [4], and our results show that this as an impact on predictions (improvements by >1% increases).

Results “per species” are available but not discussed in this report, see PDF files in outputs/figures/STOC-OED/

1. Ovaskainen O, Tikhonov G, Norberg A, Blanchet FG, Duan L, Dunson D, et al. How to make more out of community data? A conceptual framework and its implementation as models and software. *Ecology Letters* [Internet]. 2017;20(5):561–576. Available from: <https://doi.org/10.1111/ele.12757>
2. Tikhonov G, Opedal ØH, Abrego N, Lehtikoinen A, Jonge MMJ de, Oksanen J, et al. Joint species distribution modelling with the r-package Hmsc. *Methods in Ecology and Evolution* [Internet]. 2020 [cited 2020 Nov 23];11(3):442–447. Available from: <https://besjournals.onlinelibrary.wiley.com/doi/abs/10.1111/2041-210X.13345>
3. Vallé C, Poggiato G, Thuiller W, Jiguet F, Princé K, Le Viol I. Species associations in joint species distribution models: From missing variables to conditional predictions. *Journal of Biogeography* [Internet]. 2024 Feb [cited 2026 June 12];51(2):311–324. Available from: <https://onlinelibrary.wiley.com/doi/10.1111/jbi.14752>
4. Mondain-Monval T, Pocock M, Rolph S, August T, Wright E, Jarvis S. Adaptive sampling by citizen scientists improves species distribution model performance: A simulation study. *Methods in Ecology and Evolution* [Internet]. 2024 July [cited 2026 May 22];15(7):1206–1220. Available from: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.14355>